

Computers in Human Behavior

When personalization deters delegation to algorithms: Evidence from a preregistered experiment --Manuscript Draft--

Manuscript Number:	
Article Type:	Full Length Article
Section/Category:	Full Length Article
Keywords:	Delegation to AI; personalization; algorithm appreciation; algorithm aversion; Trust in AI; financial decision-making
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Abstract:	Algorithmic decision aids increasingly act on users' behalf, and in many instances users can either retain control or delegate decisions to an algorithm altogether. We test whether preference-based personalization increases or decreases willingness to delegate compared with an objective optimization criterion. In a preregistered between-subjects experiment (N = 233), participants repeatedly chose whether to decide themselves or to delegate to an algorithm. In one treatment, the delegation option is an optimization-based algorithm selecting the option with the highest expected value. In the other treatment, it is a preference-based (personalized) algorithm selecting according to participants' previously elicited preferences. We find no evidence that preference-based personalization increases delegation relative to expected-value optimization throughout the task. Instead, personalization deters delegation in two ways. First, at first contact participants are significantly less likely to delegate to the preference-based algorithm than to the optimization-based algorithm; in subsequent decisions, delegation rates are statistically indistinguishable across algorithm types. Second, categorical non-delegation (never delegating) is substantially more prevalent under preference-based personalization (15.7%) than under objective optimization (3.4%). All in all, our results are consistent with the idea that preference-matching can raise initial hesitation and heighten concerns about personal autonomy and decision control for a subset of users, functioning as an acceptance barrier even when personalization is normatively appealing.
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To the Editors
Computers in Human Behavior

29 June 2026

Dear Editors,

We are pleased to submit our manuscript, "When personalization deters delegation to algorithms: Evidence from a preregistered experiment," for consideration for publication in Computers in Human Behavior.

Algorithmic decision aids increasingly act on users' behalf, and in many everyday and high-stakes settings users can either retain control or delegate a decision to an algorithm. A widely held assumption among designers is that personalizing a system to a user's own preferences should increase its acceptance. Our paper provides a direct behavioral test of that assumption and finds that, contrary to the conventional view, preference-based personalization can deter delegation rather than encourage it.

In a preregistered between-subjects experiment ($N = 233$), participants repeatedly chose whether to decide themselves or delegate to an assigned algorithm. In one condition the algorithm followed an objective expected-value-maximizing rule; in the other it was personalized, selecting according to participants' previously elicited preferences. We find that at first contact participants were significantly less willing to delegate to the personalized algorithm than to the objective one, and that categorical non-delegation (never delegating) was substantially more prevalent under personalization (15.7% vs. 3.4%). Importantly, this resistance did not stem from doubts about the personalized algorithm's accuracy: participants expected it to match their preferences better than the objective rule, yet were no more willing to delegate to it. Our evidence instead points to a desire to retain choice authority as the relevant barrier.

We believe this work fits the scope and readership of Computers in Human Behavior for several reasons. First, it speaks directly to the human side of human-algorithm interaction, isolating how the perceived nature of an algorithm's decision rule, rather than its objective performance, shapes adoption behavior. Second, to our knowledge it is the first study to experimentally isolate the effect of the decision rule itself (preference-based versus objective) on actual delegation behavior, rather than varying interface, control, or communication format. Third, the findings have concrete design implications for robo-advisors, healthcare decision aids, and other recommender systems that routinely choose between objective optimization and preference-based personalization.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The study was preregistered (<https://aspredicted.org/7un5qf.pdf>) and received ethics approval from the Institutional Review Board of the Gesellschaft fuer experimentelle Wirtschaftsforschung e.V. (certificate no. 19HBJNwQ). All participants provided informed consent. We declare no conflicts of interest. We both have read and approved the manuscript and agree to its submission. Upon

acceptance, the data, analysis code, and study materials will be deposited in a public repository. Our use of generative AI tools in preparing the manuscript is disclosed in the manuscript itself.

We would be grateful for the opportunity to have our work considered, and we thank you for your time and consideration.

Sincerely,

Irenaeus Wolff (corresponding author), Thurgau Institute of Economics (TWI) /
University of Konstanz
Lara Suraci, DG Cities

When personalization deters delegation to algorithms: Evidence from a preregistered experiment

- Personalization deters delegation to algorithms rather than increasing it
- At first contact, delegation is lower for preference-based than EV algorithms
- Categorical non-delegation is far more common under personalization (16% vs 3%)
- Never-delegators trust the algorithm to choose well but want to keep choice authority
- Effect is concentrated at first contact, feedback-effects make it go away

WHEN PERSONALIZATION DETERS DELEGATION TO ALGORITHMS: EVIDENCE FROM A PREREGISTERED EXPERIMENT*

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This version: 29th June, 2026

Abstract

Algorithmic decision aids increasingly act on users' behalf, and in many instances users can either retain control or delegate decisions to an algorithm altogether. We test whether preference-based personalization increases or decreases willingness to delegate compared with an objective optimization criterion. In a preregistered between-subjects experiment ($N = 233$), participants repeatedly chose whether to decide themselves or to delegate to an algorithm. In one treatment, the delegation option is an optimization-based algorithm selecting the option with the highest expected value. In the other treatment, it is a preference-based (personalized) algorithm selecting according to participants' previously elicited preferences. We find no evidence that preference-based personalization increases delegation relative to expected-value optimization throughout the task. Instead, personalization deters delegation in two ways. First, at first contact participants are significantly less likely to delegate to the preference-based algorithm than to the optimization-based algorithm; in subsequent decisions, delegation rates are statistically indistinguishable across algorithm types. Second, categorical non-delegation (never delegating) is substantially more prevalent under preference-based personalization (15.7%) than under objective optimization (3.4%). All in all, our results are consistent with the idea that preference-matching can raise initial hesitation and heighten concerns about personal autonomy and decision control for a subset of users, functioning as an acceptance barrier even when personalization is normatively appealing.

Keywords: AI delegation, human-AI delegation, personalization, algorithm appreciation, algorithm aversion, trust in AI, financial decision-making, preregistered experiment.

*We are grateful to all the many people who helped us write this paper, first and foremost Chris Starmer and Robin Cubitt, to whom we owe several aspects of the design, as well as the lively research group at the TWI and participants of numerous workshops and conferences.

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Disclosure Statement

The authors declare that they have no relevant or material financial interests that relate to the research described in this paper. Funding by the Thurgovian Foundation for Science and Research is gratefully acknowledged.

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1 Introduction

Algorithmic systems increasingly make and implement decisions on users’ behalf in both everyday and high-stakes contexts, from automated driving to autonomous trading and smart-home control. Public perceptions of algorithmic decision-making vary substantially across domains and are shaped by perceived fairness, accountability, privacy, trust, and usefulness (Aysolmaz et al., 2023). In many such settings, users face a concrete choice: they can retain control and decide themselves, or delegate the decision to an algorithm. Delegation can reduce cognitive effort and simplify complex choice tasks, but it can also reshape perceived agency and responsibility for outcomes. Recent literature shows that people do delegate consequential choices to AI systems, and that this delegation depends on the decision context (Candrian and Scherer, 2022). Note that throughout the paper, we refer to these systems interchangeably as algorithms, algorithmic advice, and algorithmic decision aids. This reflects the practical reality that, from a user’s perspective, the boundary between AI and algorithmic decision rules is often opaque, and particularly so when we think about personalized systems.

The idea of personalizing algorithmic systems is to tailor content, products, and services to individual user preferences, which in turn should increase acceptance. Specifically, if individuals differ in preferences, then a decision aid that incorporates those preferences should, in principle, deliver higher subjective utility than a one-size-fits-all rule. In financial decision support, for example, systems often elicit risk preferences and translate them into portfolio recommendations. The key empirical question, however, is not whether personalization sounds attractive in principle, but whether it actually increases willingness to delegate.

Existing evidence leaves that question open. In robo-advisor research, much of the evidence focuses on what predicts whether people adopt robo-advisors and how they use them (for example, Mahmud et al., 2025), rather than on direct contrasts between preference-based personalization and objective delegation rules. Adjacent work has shown that conversational onboarding and autonomy-preserving designs shape trust and uptake (Hildebrand and Bergner, 2021; Fink et al., 2024). However, these studies vary the interface, degree of user control, or communication format—not the nature of the underlying decision rule itself. To our knowledge, no study has experimentally tested whether making an algorithm’s decision rule preference-based, rather than objectively optimizing, changes actual delegation behavior. More generally, behavioral evidence shows that delegation to algorithms is shaped by more than expected performance: people sometimes avoid algorithms and sometimes prefer them based on the context (Dietvorst et al., 2018; Logg et al., 2019); they need to understand how

recommendations are generated before accepting them (Yeomans et al., 2019); and acceptance depends on moderators such as perceived control and task features (Mahmud et al., 2022; Chugunova and Sele, 2022).

Preference-based personalization can therefore be double-edged. While it may increase perceived fit, it may also encroach on users’ sense of choice authority: a system that claims to decide *as you would* substitutes for personal judgment more directly than an impersonal optimization rule, making delegation feel like ceding active control rather than outsourcing a computation. Our own post-task evidence is consistent with this interpretation: many categorical non-delegators explicitly report wanting to keep responsibility for the choice, suggesting that the barrier is not fear of being held responsible for a bad choice or outcome, but rather a positive valuation of exercising choice authority. This reading aligns with evidence that perceived control over one’s choices—not just expected benefits—is a key driver of algorithm acceptance (Schaap et al., 2024). It also connects to findings that algorithmic transparency can paradoxically heighten resistance: disclosing how a system decides can trigger negative social interpretations (Chen et al., 2024) and induce pushback in algorithmic management contexts (Hu et al., 2024), suggesting that transparency about a system’s decision basis may itself contribute to a heightened sense of reduced agency.

These issues are especially salient at first contact, when users have no experiential basis for evaluating an algorithm and must judge based on abstract cues about its decision logic. Stable preferences for decision autonomy can constrain delegation even when delegation is instrumentally attractive (Ivanova-Stenzel and Tolksdorf, 2025), making the perceived loss of choice authority especially salient at the moment of delegation. Task characteristics matter too: trust and reliance depend on perceived objectivity and time pressure (Yang et al., 2025). Skepticism does not only operate before first contact, however: once delegation has begun, reactions to algorithmic errors are an important driver of whether users keep relying on a system. A robust finding is that people abandon algorithms more readily than they switch away from doing tasks themselves or delegating to another human after seeing comparable mistakes: algorithmic errors are, in effect, forgiven less (Dietvorst et al., 2015; Renier et al., 2021). We will see that this asymmetry shapes how participants respond to feedback in our setting as well. Together, these factors suggest substantial skepticism about delegating to algorithms, both at the point of adoption and in the face of subsequent errors.

To understand what shapes these first-contact decisions, we must study observable behavior rather than attitudes alone. Behavioral reliance and self-reported trust need not coincide; evidence from AI-advisor interactions shows that determinants of behavioral trust can diverge from self-reported trust (Schreibelmayr and Mara, 2025). Moreover, even in repeated interaction settings, the mere framing of advice as AI-generated can produce systematic overreliance (Klingbeil et al., 2024),

underscoring that label-driven cues persist beyond first contact. This motivates our focus on observable delegation behavior, treating first contact as a critical moment when abstract cues about an algorithm’s decision logic are especially influential in shaping whether users delegate.

In this study, we contrast two decision rules that are both plausible in financial choice settings but may elicit different reactions. One algorithm follows an *objective optimization rule* and selects the option with the highest expected value (EV), which may appear impersonal and easier to justify as a rule-based benchmark. The other is *preference-based personalization* in a narrow sense: it predicts a participant’s preferred choice from their previously elicited preferences rather than maximizing EV. In implementation terms, the personalized algorithm maps each participant’s earlier choices in a simpler elicitation environment to corresponding choices in a more complex environment later in the experiment; it does not involve adaptive updating or learning during the delegation task.

To test this contrast, we run a preregistered between-subjects experiment with repeated delegation decisions. Participants first complete 12 binary lottery choices (Part 1), which provide the preference-elicitation input for the personalized algorithm. They then enter Part 2, where in each round they choose either to decide themselves among eight lottery options or to delegate that choice to the assigned algorithm (EV or preference-based, depending on the treatment).

This design allows us to isolate a preregistered first-contact test, before any outcome feedback is observed, and to distinguish it from behavior after feedback has accumulated. The comparison between our algorithms connects to evidence that responsibility-sharing considerations can shape advisor choice in human-versus-algorithm settings (Gazit et al., 2023) and to recent studies of financial delegation and outcome evaluation (Ismagilova and Ploner, 2025). Related findings in financial delegation are also suggestive of responsibility-sharing considerations (Werner et al., 2026), even though evidence on perceived responsibility differences between human and machine partners is mixed (Kirchkamp and Strobel, 2019).

Preregistered hypothesis and analysis plan

Consistent with our preregistration (<https://aspredicted.org/7un5qf.pdf>), our main hypothesis is that the diminished sense of choice authority from delegating to a personalized algorithm outweighs the benefits it renders in terms of better-fitting choices:¹

H 1. People are more likely to delegate financial decisions to an EV-maximizing algorithm than to a personalized algorithm that incorporates their (revealed)

¹The direction of the prediction was informed by a pilot study with a different design (reported in the second author’s PhD thesis); see the Discussion section for a comparison of results.

preferences.

Our preregistered primary analyses test this hypothesis in two complementary ways. First, we test the treatment difference in delegation in the first decision (round 1) using a one-sided Boschloo exact test. This isolates first contact, when participants have no outcome feedback and must decide whether to delegate based on their interpretation of the algorithm. Second, we test whether EV maximization leads to higher delegation overall by comparing participants’ average delegation frequencies across rounds using a one-sided Wilcoxon–Mann–Whitney test.

To account for the repeated-decision structure and additional determinants of delegation, our preregistration also specifies a multivariate regression analysis. We estimate a model with treatment indicators interacted with the decision period, decision-ID fixed effects, and participant-clustered standard errors. We further include the feedback history in the form of including the share of zero outcomes among prior delegated choices, to control for how experienced payoffs may influence subsequent delegation decisions. This preregistered specification tests whether any treatment difference varies systematically over decisions while controlling for option-set heterogeneity and participants’ feedback histories.

Study overview and supplementary analyses

We implement our design in a between-subjects experiment in which participants repeatedly choose whether to decide themselves or delegate to the assigned algorithm. The repeated-decision setting provides a direct behavioral measure of delegation and allows us to distinguish first-contact delegation from later decisions.

In addition to the preregistered tests, we report a small set of analyses to characterize heterogeneity in delegation propensity. In particular, we examine categorical non-delegation (never delegating across rounds). This outcome captures a distinct form of resistance to algorithmic control transfer and is relevant for understanding which users may opt out of decision aids altogether.

In sum, this paper provides a preregistered test of a simple but consequential claim: even in a setting where preference-based personalization may appear normatively attractive, people may nonetheless be more willing to delegate to a ‘more objective’ algorithm. The contribution is threefold: we provide a preregistered behavioral comparison of objective versus personalized decision rules, we identify where the treatment effect is concentrated (first contact), and we document categorical non-delegation as a distinct uptake barrier under personalization. Together, these findings sharpen the design implication that uptake depends on perceived rule characteristics, not only on expected performance.

2 Related Work

Our work sits at the intersection of literatures on (i) the behavioral uptake of algorithmic decision aids, (ii) delegation, agency, and responsibility in human–algorithm interaction, (iii) transparency as an interpretive cue for acceptance, (iv) personalization as a design feature of algorithmic decision rules, and (v) heterogeneity in trust and delegation. Below, we situate our study within these streams, identify the specific gap our experiment addresses—the lack of a direct behavioral comparison between personalized and objective decision rules—and connect them to the robo-advisor application domain.

2.1 Algorithm aversion, appreciation, and conditions for uptake

Behavioral uptake of algorithmic advice is not one-directional: studies document both *algorithm aversion* and *algorithm appreciation*, depending on context, feedback, and comparison benchmark (Dietvorst et al., 2018; Logg et al., 2019). A core implication is that reliance cannot be predicted based on objective performance alone. Users need to make sense of how recommendations are generated before accepting them, and understandability can gate uptake independently of accuracy (Yeomans et al., 2019). Systematic evidence further shows that uptake is moderated by factors at multiple levels (algorithm, individual, and task), including perceived control and decision context (Mahmud et al., 2022; Chugunova and Sele, 2022).

This mixed-evidence baseline motivates our design choice to study delegation as an observed behavioral outcome and to isolate one concrete determinant of uptake: the algorithmic *decision rule* (objective EV maximization versus preference-based personalization).

2.2 Delegation, agency, and responsibility sharing

Delegation goes beyond advice-taking: it is a transfer-of-control decision that entails ceding choice authority to the algorithm (Candrian and Scherer, 2022). For that reason, agency and responsibility become central concerns. In advisor-choice settings, people evaluate more than expected payoff, including whether responsibility for consequences can be shared with or offloaded onto the advisor (Gazit et al., 2023). Related evidence from financial delegation suggests that delegation patterns are consistent with responsibility-sharing motivations, particularly in loss contexts (Werner et al., 2026). Yet, evidence on whether perceived responsibility differs between human and machine partners is mixed, with Kirchkamp and Strobel (2019) finding little difference in perceived responsibility or guilt.

This control-and-responsibility perspective is consistent with adjacent evidence: perceived trustworthiness dimensions shape acceptance of algorithmic decision-makers (Höddinghaus et al., 2021); preserving human involvement can increase uptake but may lower accuracy (Sele and Chugunova, 2024); and stable preferences for decision autonomy can constrain delegation even when delegation is instrumentally attractive (Ivanova-Stenzel and Tolksdorf, 2025). Taken together, this evidence implies that delegation is shaped by perceived control and responsibility allocation, not only expected returns. This motivates our focus on observed delegation behavior and makes the EV-versus-personalized comparison directly informative, because the two rules differ in how strongly they cue impersonal objectivity versus personal ownership of outcomes.

2.3 Transparency, and interpretive cues for acceptance

Personalization and transparency are often presented as acceptance-enhancing design features, but recent evidence suggests they can also trigger resistance. In recommendation contexts, greater transparency can backfire when users infer socially negative meanings (for example, feeling judged or dehumanized), reducing acceptance (Chen et al., 2024). In algorithmic management, transparency can likewise show a non-monotonic effect, where additional transparency eventually increases resistance rather than compliance (Hu et al., 2024). Related evidence further indicates that acceptance is driven less by whether the decision-maker is labeled algorithmic or human, and more by expected benefits and perceived control (Schaap et al., 2024).

The broader explainability literature is consistent with this interpretation: transparency and explanations matter, but their effects depend on what they change psychologically. Fairness/accountability/transparency perceptions shape trust and adoption (Shin and Park, 2019); explanations can improve trust calibration and behavior in some settings (Leichtmann et al., 2023); and combining explanation with decision control can improve compliance, though design complexity can also offset those gains (Westphal et al., 2023). In applied decision-support experiments, transparency influences advice use via trusting beliefs and perceived understanding, with effects that depend on the information environment and experienced system performance (Ning et al., 2024; Sun et al., 2026). For our setting, this implies that users’ interpretation of decision logic can shape delegation independently of objective expected returns.

2.4 Decision-rule comparison gap: personalized versus objective rules

Despite the breadth of the literatures reviewed above, a striking gap remains: to our knowledge, no study has experimentally tested whether making an algorithm’s decision rule *preference-based* (as opposed to following an impersonal, objective criterion) changes actual delegation behavior. Several lines of work come close but address different questions.

First, the “algorithm modification” paradigm pioneered by Dietvorst et al. (2018) shows that giving users even slight control over an algorithm’s output increases adoption. This demonstrates that *user input* matters, but the manipulation concerns users’ ability to adjust the output, not whether the underlying decision rule incorporates their preferences or maximizes an objective criterion.

Second, the robo-advisor design literature examines how features such as conversational onboarding (Hildebrand and Bergner, 2021), user controllability over rebalancing (Kim et al., 2025), and the performance–control trade-off in configuration choices (Rühr, 2020) shape adoption and trust. These studies vary the *interface* and the *degree of control*, but hold the nature of the decision rule itself constant or leave it as an implicit design feature.

Third, theoretical and empirical work on robo-advisor personalization has focused on optimizing portfolio allocation through more granular client interaction (Capponi et al., 2022) or on comparing algorithm-selected portfolios with self-directed choices (Boulu-Reshef et al., 2026). These contributions are informative about *performance* implications of personalization, but do not ask whether users are more or less willing to delegate when the algorithm’s rule is personalized.

This gap is consequential beyond academic interest. In practice, financial robo-advisors, healthcare decision aids, and algorithmic recommender systems routinely face the design choice between objective optimization criteria (such as expected-value or evidence-based guidelines) and preference-based rules that incorporate individual user profiles. If personalization reduces rather than increases the willingness to delegate—as our evidence suggests—then system designers may inadvertently raise adoption barriers by prioritizing preference-matching. Understanding this trade-off is directly relevant for any domain where algorithmic decision aids must be voluntarily adopted to be effective.

2.5 Heterogeneity and contextual moderators of trust and delegation

Recent evidence shows that algorithm uptake is heterogeneous across people, tasks, and decision environments (Ferraz et al., 2025; Yang et al., 2025). At the individual level, delegation varies with psychological readiness and other personal

traits linked to delegation behavior (Kim, 2026; Ferraz et al., 2025). Beyond stable traits, uptake also depends on decision architecture: autonomy-preserving choice designs can increase recommendation acceptance (Fink et al., 2024). At the task level, uptake depends on objectivity and time pressure (Yang et al., 2025), while overreliance risks remain salient when users follow advice against their own information (Klingbeil et al., 2024). Taken together, these cross-level moderators imply that average delegation rates can mask meaningful subgroup patterns. For our study, this motivates looking beyond means and explicitly reporting categorical non-delegation as a distinct acceptance pattern.

2.6 Financial advice and robo-advisors as an application domain

Although our task is experimentally stylized, it maps naturally to financial delegation settings where users decide whether to retain control or hand choice authority to an algorithmic decision aid. This is exactly the context in which objective optimization and preference-based personalization often coexist in practice. Existing robo-advisor evidence is informative but fragmented: studies document adoption patterns and how users combine advisor recommendations with their own judgment (Figà-Talamanca et al., 2022; Mahmud et al., 2025), outcome bias in the evaluation of delegated financial decisions, whether made by a human or an algorithm (Ismagilova and Ploner, 2025), and potential performance gains from robo-advice relative to self-directed choices (Lambrecht et al., 2023).

This makes financial decision-making a strong test bed for our question: whether changing the algorithmic decision rule changes willingness to delegate, even when both rules are plausible.

2.7 Synthesis and contribution

Across these literatures, the common message is that delegation to algorithms is jointly shaped by decision-rule design, perceived control and responsibility, trust calibration, and context-dependent interpretation of transparency and autonomy-related design cues (Dietvorst et al., 2018; Chugunova and Sele, 2022; Schaap et al., 2024; Fink et al., 2024; Hu et al., 2024). Yet as we have documented above, the existing evidence base contains a notable gap: while adjacent work has studied algorithm modification, controllability, conversational design, and performance implications of personalized portfolios, no study has experimentally isolated the effect of the decision rule itself—specifically, whether making the rule preference-based rather than objectively optimizing changes users’ willingness to delegate. This gap matters practically because robo-advisors, healthcare decision aids, and algorithmic recommender systems routinely face exactly this design choice, and a

wrong assumption about the direction of the effect could raise rather than lower adoption barriers.

Our study addresses this gap with a preregistered behavioral test in a repeated-decision delegation paradigm, contrasting an EV-maximizing rule with a preference-based rule. By jointly examining first-contact delegation, average delegation across rounds, and categorical non-delegation, we connect classic aversion/ appreciation evidence with newer findings on autonomy, transparency backfire, and responsibility attribution (Logg et al., 2019; Fink et al., 2024; Hu et al., 2024; Gazit et al., 2023).

3 Method

3.1 Participants and preregistered exclusions

Participants were recruited via Prolific from a UK-based participant pool and completed the study online. The preregistered target was 120 participants per treatment condition before exclusions. We recruited $N = 240$ participants and then applied preregistered exclusion rules for data quality and protocol compliance. Exclusions covered failed bot/slider checks at first attempt, failed comprehension checks after the second attempt, failure to confirm careful participation, disclosed AI usage during the study, implausibly short completion times under the preregistered log-time rule, and incomplete data. The final analysis sample is $N = 233$ (EV condition: $n = 118$; preference-based condition: $n = 115$).

3.2 Design and task

The study used a between-subjects design with two algorithm conditions. In both conditions, participants completed a two-part experiment programmed in the online-ready version of z-Tree (version 6.0.0 beta 69, Fischbacher, 2007).

Part 1 consisted of 12 binary lottery choices used to elicit revealed risk preferences. Eight of these tasks were directly linked to Part 2 (see below); the remaining four served as filler tasks, as one of several measures to prevent participants from anticipating the structural correspondence between Part 1 and Part 2. Participants were truthfully told that “[t]he questions in this part are **hypothetical**, but the choices **may be meaningful for Part 2** (recall that your bonus payment from the experiment depends on what happens in Part 2).” The highlighted words were boldfaced in the original instructions.

We deliberately designed Part 1 to give participants every opportunity to make a well-considered, informed choice, precisely because these are the choices on which the preference-based algorithm is based. To this end, three features

distinguish Part 1 from the later delegation task. First, unlike the time-constrained decisions in Part 2 (see below), participants faced no time limit in Part 1 and could deliberate for as long as they wished. Second, each Part-1 task presented a clean choice between only two lotteries, rather than the eight options participants would later face in Part 2. Third, payoffs were matched to states so that they would appear in decreasing order to ease comparison, as can be seen in the upper panel of Figure 1. In addition, we asked participants after each task in Part 1 how confident they were in their choice.

	State 1	State 2	State 3	State 4	State 5	State 6	State 7	State 8	State 9	State 10
Option 1	225	225	225	225	225	225	225	205	0	0
Option 2	525	525	525	525	525	0	0	0	0	0

	State: 1	2	3	4	5	6	7	8	9	10
Option 1	225	225	0	225	0	225	225	225	225	0
Option 2	525	0	505	0	525	0	0	0	525	505
Option 3	525	0	525	0	525	0	0	0	525	525
Option 4	225	225	0	225	205	205	205	225	225	0
Option 5	225	205	0	205	205	225	225	205	225	0
Option 6	225	225	0	225	205	225	225	225	225	0
Option 7	525	0	525	0	525	0	0	0	225	525
Option 8	505	0	525	0	505	0	0	0	505	525

Figure 1: Example screens from Part 1 (preference elicitation, upper panel) and Part 2 (delegation decisions with lottery choices, lower panel).

In Part 2, participants completed eight incentivized decisions between eight lottery options each, as exemplified in the lower panel of Figure 1. These eight Part-2 tasks were derived from eight mapped Part-1 binary tasks (all but the four filler tasks). Specifically, each mapped Part-1 task provided two “anchor” options; for Part 2, we first transformed each anchor option by permuting payoffs across states (Option 6 and Option 3 in the lower panel of Figure 1), and then added three decoy options for each anchor option. Each added option was state-wise dominated by one of the anchor options (i.e., worse in at least one outcome and no better in any other), without making the dominance relationship immediately obvious from the displayed table. This yields eight options per Part-2 task (two transformed anchors plus six decoy options). The within-task display order of the eight options was set using a fixed pre-specified rule rather than per-task randomization. We did this because unconstrained randomization would occasionally produce display orders where the three decoy options belonging to the same anchor would appear consecutively next to the anchor option itself.

This is something we wanted to avoid, as such a clustering could have alerted attentive participants to the underlying construction logic of Part-2 tasks. The fixed ordering rule was designed to interleave anchor-based groups so that no such visually suspicious pattern would arise. Note also that the ordering rule was different between option sets.

The order of the 12 Part-1 tasks was randomized with one pre-defined constraint: positions 11 and 12 were occupied by two (randomly selected) tasks from the four Part-1 filler tasks that had no mapped counterpart in Part 2. Consequently, all eight mapped Part-1 tasks appeared within the first ten Part-1 positions, while still preserving randomization of task order subject to this constraint.

The procedure for each round of Part 2 was as follows: Participants saw the full table for five seconds. Then, they saw a different screen on which they had five seconds to choose between delegating to the algorithm corresponding to their treatment condition and choosing an option themselves. In the EV condition, delegation selected the lottery with the highest expected value. In the preference-based condition, delegation selected the non-dominated lottery which the participant had preferred in Part 1 over the other non-dominated lottery from the current task. Importantly, in both treatments the delegated option was always one of the two transformed anchor options and therefore never one of the six decoy options. Assignment to condition was randomized at the participant level. In the EV condition, participants were informed that “[the] algorithm works out how much each option would yield on average, and selects the option for which this is highest.” In the preference-based condition, they instead read that “[the] algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.”

If they did not delegate, they saw the table containing the eight options for up to eight more seconds, during which they had to choose one of the options. After each Part-2 decision, feedback provided full resolution of uncertainty: participants were shown the chosen option and the realized payoff from one randomly drawn state of the world for that round. If they did not make a choice in the payoff-relevant task, they did not earn a bonus payment from the experiment. The total payoff from the experiment was GBP 2.00 plus the points earned in one randomly selected round of Part 2, converted at a rate of 1 penny per point (participants could earn up to GBP 8.90 in addition to the base payoff). On average, participants earned GBP 4.14 including the base payoff for a median time of just under 16 minutes.

3.3 Procedure

After being welcomed to the study, participants saw a symbolic representation of a bicycle (two red tires and two triangles in yellow and green that together formed

the frame), being asked to indicate what that figure represented (a banana, a traffic light, darts games, a bicycle, a robot, or ‘nothing, it’s randomly placed shapes’). At the time, the agentic AIs would typically choose the traffic light, presumably because of the colouring of the shapes making up the bicycle. Following that, participants would read a broad overview of upcoming tasks (‘In Parts 1 and 2: You participate in different decision-making tasks. In Part 3: You answer a short questionnaire’, followed by the information that they would receive GBP 2 for participating and up to GBP 8.90 as a bonus, depending on decisions and on chance). Then, they would have to give consent to participating and to our using of their data for the purposes of this publication, which was followed by a final attention and ‘system’ check (‘Please select “Disagree” on the following question’ and ‘please click on the white line in such a way that the appearing blue dot is close to a value of 50.’).²

After these checks, Part 1 (preference elicitation) started, followed by a question on how satisfied they were with their choices from Part 1 overall. Then, participants completed Part 2 (delegation decisions with round-by-round outcome feedback), followed by a question of whether they would delegate to the algorithm in case they had to choose to delegate or not for 10 (hypothetical) additional tasks. After indicating that choice, participants estimated (on a 0–10 integer scale) in how many of those 10 tasks the algorithm would choose a non-preferred option and in how many it would choose a “completely weird” option. Before the final post-task questionnaire containing demographics and attitudinal items, we asked them for self-reported reasons for their choice in the hypothetical choice at the end of Part 2. Because our theoretical focus is reliance behavior, primary outcomes are behavioral rather than attitudinal.

3.4 Ethics approval

The study received ethics approval from the Institutional Review Board of the Gesellschaft für experimentelle Wirtschaftsforschung e.V. (GfeW; German Association for Experimental Economic Research), certificate no. 19HBJNwQ (status: approved; expedited review; approval date: 11/13/2025; certificate validity through 11/13/2027; see <https://gfew.de/ethik/19HBJNwQ>). All participants provided informed consent before participation.

²The ‘system check’ was meant to screen out certain types of bots, as, in an invisible box below the displayed white line, there was a ‘50’ being displayed right next to the (visible) end-of-the-scale ‘100’.

3.5 Preregistered outcomes and analyses

The preregistered hypothesis (H1) predicted higher delegation to the EV-maximizing algorithm than to the preference-based algorithm. The preregistration document is available at <https://aspredicted.org/7un5qf.pdf>.

Primary preregistered analyses were:

1. a one-sided Boschloo exact test for the treatment difference in delegation in round 1 (first contact), and
2. a one-sided Wilcoxon–Mann–Whitney test comparing participant-level average delegation rates across Part 2.

We additionally estimated a preregistered linear probability model with decision-set fixed effects and participant-clustered standard errors, including treatment-by-period interactions and the share of outcomes with a payoff of zero among prior delegated choices. We further estimated an exploratory specification excluding round 1 to separate first-contact effects from choices made after feedback experience.

4 Results

4.1 Overview

Two empirical patterns organize the results. First, treatment differences are concentrated at first contact. Second, categorical non-delegation is substantially more common under preference-based personalization.

4.2 Primary test: first-contact delegation

As a benchmark for interpreting delegation behavior, choice quality among non-delegated decisions was low in both treatments. The share of non-dominated self-chosen options was 34.1% in the EV condition and 35.1% in the preference-based condition, implying that roughly two-thirds of self-made choices were dominated. Hence, participants had a clear instrumental reason to delegate. On average, delegated decisions yielded 243.49 points (GBP 2.43), compared to 195.18 points (GBP 1.95) for non-delegated decisions, a difference of approximately 25%.

In round 1, delegation was higher in the EV condition than in the preference-based condition (66.9% vs. 55.8%). The preregistered one-sided Boschloo exact test yields $p = 0.0413$, supporting H1 for first-contact behavior. Figure 2 visualizes the round-level profile and shows that there is a clear treatment gap in the first decision.

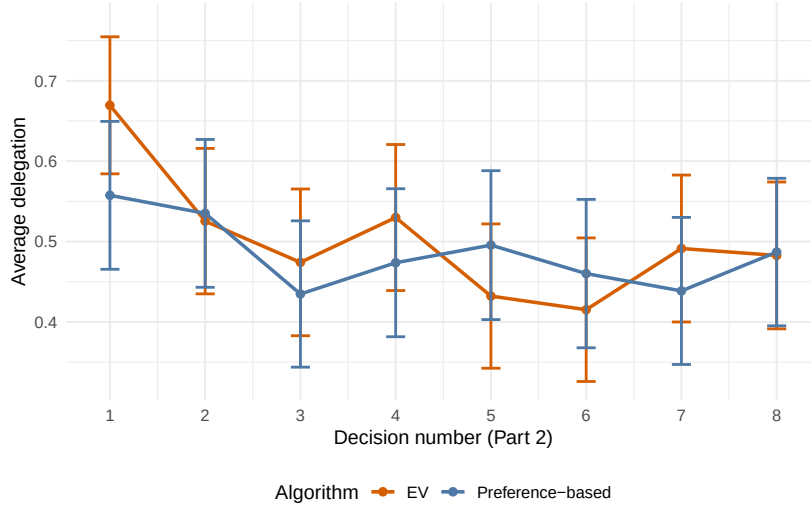


Figure 2: Average delegation over decisions in Part 2 by treatment condition. Whiskers indicate 95%-confidence bands.

4.3 Primary test: delegation across repeated decisions

At the participant level, the preregistered one-sided Wilcoxon–Mann–Whitney test on average delegation across Part 2 yields $W = 6428$ and $p = 0.2425$. We therefore do not find evidence of a treatment difference in average delegation across rounds.

Descriptively, this non-difference reflects post-first-contact convergence that is driven by a strong decline in delegation in the EV condition after round 1 (Figure 2). We treat this pattern as descriptive because our design was not optimized to identify why EV delegation appears more sensitive to early feedback after first contact. Notwithstanding, we will report on how this pattern relates to ‘wins’ (rounds in which the realized outcome was positive) and ‘losses’ (rounds in which the realized outcome was zero).

The preregistered regression specification points in the same direction. The EVAlgorithm coefficient is small and statistically insignificant ($\beta = -0.0487$, $SE = 0.0831$, $p = 0.558$). Likewise, both interaction terms are imprecisely estimated (EVAlgorithm $\times$ WithinPartDecisionNumber: $\beta = 0.0019$, $SE = 0.0115$, $p = 0.871$; EVAlgorithm $\times$ share[Payoff = 0]: $\beta = -0.0497$, $SE = 0.0977$, $p = 0.612$), indicating no robust treatment difference over repeated decisions. By contrast, the share of prior delegated outcomes with a payoff of zero is strongly negatively associated with subsequent delegation ($\beta = -0.3854$, $SE = 0.0781$, $p < 0.001$).³

³Running an analogous regression with fixed effects for the task on round-1 data only yields the same result as the Boschloo test, with $\beta = 0.113$, $SE = 0.065$, and $p = 0.041$ (one-sided).

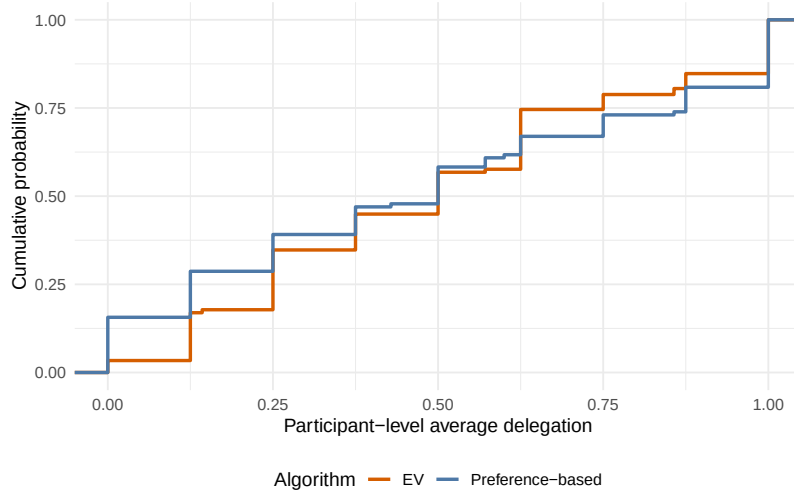


Figure 3: ECDF of participant-level average delegation by treatment condition.

Taken together, the preregistered tests indicate that the treatment effect is concentrated at first contact and does not persist over repeated decisions, while negative feedback (in particular prior zero-outcomes) is strongly associated with lower subsequent delegation. Note that the EV-algorithm will lead to more zero-outcomes (and thus, lower subsequent delegation rates) than the preference-based algorithm for the vast majority of participants (in expectation, this holds true for 89.7% of the participants), so that the vanishing of our treatment effect after period 1 is not so surprising overall. The cross-participant distribution aggregated over all rounds is shown in Figure 3. Note that the convergence in delegation behavior over rounds does not diminish the substantive relevance of first-contact effects: in many real settings, the initial adoption decision can determine whether users return to a system at all.

4.4 Categorical non-delegation

This section reports an outcome that was not preregistered (categorical non-delegation) that is complementary to the preregistered primary tests.

The second clear pattern is categorical non-delegation, which can be seen from the left end of Figure 3. In the preference-based condition, 18 of 115 participants never delegated (15.7%); in the EV condition, this applies to only 4 of 118 participants (3.4%). This gap points to a distinct acceptance barrier under personalization for a subset of users.⁴ Note that although there also seems to be a

⁴The same direction appears in an earlier study with a different design (reported in the second

treatment difference in always-delegators, this difference is far less pronounced (19.1% vs. 13.6%).

Never-delegators were not meaningfully different in age from participants who delegated at least once (mean age 41.8 vs. 42.9 years; $t = -0.329$, $p = 0.745$), and we likewise find no clear association with household income (categorical; $\chi^2 = 13.171$, $df = 11$, $p = 0.282$) or education (categorical; $\chi^2 = 1.859$, $df = 5$, $p = 0.868$), and both spent similar amounts of time in the experiment (mean number of seconds 1113 for delegators vs. 1237 for never-delegators, $t = -0.971$, $p = 0.341$). And while there is a tendency for male participants to be never-delegators more often than females (12.0% vs. 6.3%), the evidence is weak (Boschloo exact test, $p = 0.156$). We also do not find meaningful differences in self-reported willingness to take financial risks (4.53 vs. 4.68; $t = -0.224$, $p = 0.825$), self-reported response reliability/care (6.91 vs. 6.91; $t = -0.059$, $p = 0.954$), or frequency of AI use/interactions (categorical: 5.87 vs. 5.18; $t = 0.866$, $p = 0.395$). Within the preference-based condition, never-delegators actually reported slightly higher satisfaction with their Part-1 choices than delegators (7.94 vs. 7.56; $t = -1.400$, $p = 0.169$). This suggests that categorical non-delegation in this condition does not originate from dissatisfaction with the preference profile that served as the algorithm’s input.

4.5 Responses to feedback

This preregistered secondary analysis characterizes behavioral dynamics rather than serving as an additional primary test of H1.

To characterize feedback dynamics, we estimate a win-stay/lose-shift model on rounds 2–8, regressing an indicator for staying with the same delegation/non-delegation choice on a prior win (a previous-round payoff larger than zero), with task and decision-number fixed effects and participant-clustered standard errors. A prior win is strongly associated with staying ($\beta_{\text{lagWin}} = 0.169$, $SE = 0.035$, $p < 0.001$), and this win-stay/lose-shift tendency is statistically indistinguishable across treatments (EValgorithm $\times$ lagWin: $\beta = 0.009$, $SE = 0.049$, $p = 0.857$). Participants in the EV condition stay somewhat less overall ($\beta_{\text{EValgorithm}} = -0.105$, $SE = 0.050$, $p = 0.036$). In other words, both treatments display the same basic pattern: after a loss, participants are more likely to switch their delegation decision than after a win.

A more revealing picture emerges when we distinguish *who* produced the prior outcome—that is, whether the previous round was delegated (so that “the algorithm lost or won”) or self-chosen. Estimating the model separately by

author’s PhD thesis): 32/189 never-delegators in preference-based treatments versus 17/187 in EV treatments (one-sided Boschloo exact test, $p = 0.0135$).

treatment and interacting a prior loss with prior delegation, the prior-loss $\times$ prior-delegation interaction is negative and significant in both conditions (EV: $\beta = -0.185$, $SE = 0.070$, $p = 0.009$; preference-based: $\beta = -0.140$, $SE = 0.064$, $p = 0.030$). That is, a loss prompts more switching in general, but a loss that the algorithm produced prompts substantially more switching than a self-inflicted loss. This is visible in the raw stay rates: after a delegated loss, the stay rate is the lowest in both conditions (54.7% in the preference-based and 44.4% in the EV condition), well below the stay rate after a delegated win (78.5% and 69.7%) and even below that after a self-chosen loss (67.7% and 58.3%).

This pattern connects to the broader finding that algorithmic mistakes tend to be forgiven less readily than one’s own (Dietvorst et al., 2018; Renier et al., 2021): participants abandon delegation disproportionately after the algorithm delivers a bad outcome, even though delegated choices were on average substantially more profitable. The dynamics are therefore similar across treatments and reinforce the broader message that the treatment difference is concentrated at first contact, rather than in how participants respond to feedback thereafter.

4.6 Exploratory mechanisms: expectations and post-task reasons

To better understand the behavioral patterns above, we draw on two sets of exploratory measures collected after Part 2: participants’ quantitative expectations about how the algorithm would perform, and their self-reported reasons for a hypothetical further delegation choice. We organize the discussion around three questions: (i) Did participants believe the preference-based algorithm actually captured their preferences? (ii) Which expectations are most tightly linked to delegation, and does this differ by algorithm type? (iii) What do the open- and closed-item reasons reveal about the motives of those who delegate sometimes or always versus those who never do?

Did participants trust the personalized algorithm to match their preferences?

A natural concern about our design is that lower delegation under personalization might simply reflect participants doubting that the preference-based algorithm captured their preferences in the first place. The expectation data speak directly against this concern. After the incentivized rounds, participants estimated, out of 10 hypothetical further tasks, in how many tasks the algorithm would pick a non-preferred option and in how many it would pick a “completely weird” option. Participants in the preference-based condition expected the algorithm to make markedly *fewer* non-preferred choices than those in the EV condition (on average

3.78 vs. 5.00 out of 10; $t = -4.15$, $p < 0.001$; Wilcoxon rank-sum $p < 0.001$; Figure ??). Expectations about “completely weird” choices, in contrast, did not differ across conditions (1.57 vs. 1.78; $t = -0.94$, $p = 0.351$). Participants thus clearly credited the personalized algorithm with doing its job: relative to objective optimization, it was expected to select options aligned with their preferences substantially more often, without any rise in the perceived risk of grossly erratic choices. This holds even for the participants who opted out entirely: within the preference-based condition, never-delegators expected just as few non-preferred choices as delegators (3.83 vs. 3.77 out of 10; $t = -0.11$, $p = 0.912$). Categorical non-delegation under personalization thus cannot be attributed to a belief that the algorithm failed to capture one’s preferences, reinforcing the earlier finding that these never-delegators were, if anything, slightly *more* satisfied with the Part-1 choices that fed the algorithm. Note further that if we restrict the sample to only those participants who expect a maximum of 2 mistakes by the algorithm (21% of the participants), our treatment differences become stronger rather than weaker, adding further evidence that the treatment effects do not seem to be driven by participants not believing in the ability of the preference-based algorithm to do its job.

This makes the reduced delegation under personalization all the more striking. Even though participants believed the preference-based algorithm would more often choose in line with their preferences, they were no more—and at first contact significantly less—willing to delegate to it. The barrier therefore does not appear to lie in doubts about the algorithm’s accuracy. Consistent with this, the two explanation-demand items (“I find it unpleasant that I don’t know how exactly the computer chooses” and “I would have preferred more information about how the computer makes its decisions”) were selected only infrequently, as was the profiling-related item (“I don’t like to be analysed by an algorithm”). Participants’ reservations were thus not primarily about understanding, transparency, or competence at preference-matching, pointing instead toward concerns about ceding choice authority itself, as we will show below.

What distinguishes delegators from never-delegators?

The closed-item reasons (Figure 4) vary more by delegation type than by treatment. The only visible treatment contrast—larger never-delegator/delegator gaps in the EV condition—we do not interpret, because the EV never-delegator subgroup comprises only four participants. Two patterns stand out. First, participants who delegated at least once were far more likely to cite performance-based motives: 70.1% indicated that the computer would choose better, against only 36.4% of never-delegators (that they would choose better). Second, never-delegators more often selected reasons centered on general distrust of the algorithm (50.0% vs.

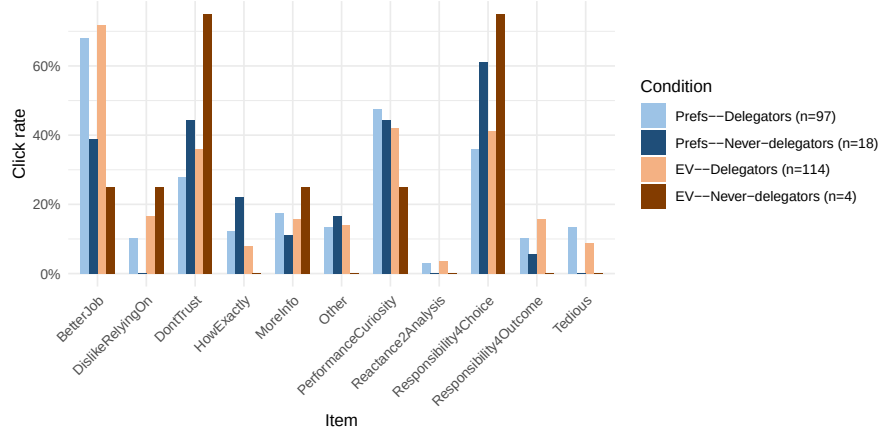


Figure 4: Distribution of reasons indicated for delegating or not 10 additional hypothetical tasks to the algorithm.

32.2%; recall, however, that those who indicate the strongest belief in that the algorithm does its job do not exhibit a smaller treatment effect; if at all, the effect is stronger in that subset) and on retaining personal responsibility for the choice (63.6% vs. 38.9%). This mirrors the treatment-level non-delegation pattern and is consistent with a subgroup that weights agency and responsibility concerns more heavily than expected performance. Among the remaining items, finding the task too tedious appears (at a low level) only among delegators (10.9% vs. 0%), consistent with an effort-saving channel, and open-text responses repeatedly mention time pressure, a general choice-friction channel that does not isolate a treatment-specific mechanism.

These mechanism patterns should be interpreted cautiously. The reason measures are ex-post and refer to a hypothetical additional choice rather than to the incentivized rounds themselves, so they function as suggestive proxies for motives rather than as causal identification of psychological channels.

5 Discussion

5.1 Core finding

This preregistered study shows that preference-based personalization can deter delegation rather than increase it. We observe this deterrence in two forms: lower delegation at first contact and a larger subgroup of participants who never delegate under personalization.

The first-contact finding is important because it captures behavior before

participants can learn from outcome feedback. At that stage, delegation is driven primarily by how participants interpret the algorithmic rule itself. Our evidence is consistent with the interpretation that an impersonal EV rule is initially easier to accept as a legitimate external computation device than a profile-based personalized rule based on the user’s own revealed preferences. Even though subsequent rounds show convergence in average delegation (driven by reduced delegation to the EV algorithm), first-contact differences remain practically important because initial adoption decisions can have path-dependent downstream effects in real-world deployment where customers may not constantly be prompted to make a delegation choice like in our experiment (think of immediate opt-out, lower willingness to return, and weaker opportunity for learning-by-use).

5.2 Interpretation and relation to prior work

Our main contribution is to show that algorithm uptake depends on the decision rule, not just on whether a system is algorithmic. The contrast between expected-value maximization and preference-based personalization indicates that users may resist personalization despite its normative appeal.

The categorical non-delegation pattern is particularly informative: many participants entirely avoid delegation under personalization, consistent with work on autonomy, responsibility, and control as drivers of acceptance. Exploratory post-task reasons (Section 4.6) align with this interpretation, especially the stronger emphasis among never-delegators on responsibility and distrust, while concerns about being algorithmically analyzed appear comparatively rare. Because these reason measures are ex-post and hypothetical, this evidence remains suggestive rather than causal. Crucially, the lower delegation under personalization does not stem from participants doubting that the personalized algorithm captured their preferences: they expected it to make significantly fewer non-preferred choices than the EV algorithm (3.78 vs. 5.00 out of 10; $p < 0.001$), while expectations about “completely weird” choices did not differ across conditions. Participants thus credited the preference-based algorithm with successfully matching their preferences, yet were no more willing—and at first contact less willing—to delegate to it. This strengthens the interpretation that the relevant barrier is the act of ceding choice authority itself rather than skepticism about the algorithm’s accuracy. A complementary channel operates through how algorithmic errors are treated once they occur: in both conditions, participants withdrew from delegation disproportionately after the algorithm produced a bad outcome (Section 4), echoing evidence that algorithmic mistakes are forgiven less readily than one’s own (Dietvorst et al., 2018; Renier et al., 2021).

These findings are consistent with an earlier study on the same question (reported as a chapter in the second author’s PhD thesis), which motivated and

informed the design of the present preregistered experiment. In that study, the delegation gap between EV and preference-based treatments was larger (about 20 percentage points) and persisted over decisions. Recomputing the categorical non-delegation contrast for that earlier sample also yields a significant difference (32/189 in preference-based treatments vs. 17/187 in EV treatments; one-sided Boschloo exact test, $p = 0.0135$).⁵

5.3 Limitations

The findings should be interpreted in light of several limitations. First, the study is an online experiment with stylized financial lotteries, so external validity to field settings should be interpreted cautiously. Second, although behavioral outcomes are well suited for studying reliance, the design does not causally identify the psychological mechanism behind first-contact hesitation or categorical non-delegation (e.g., perceived objectivity, profiling concerns, or accountability salience). Third, the present analyses focus on short-run repeated interaction; longer-run adaptation to personalized systems remains an open question.

5.4 Implications and future research

For designing algorithmic decision systems, the results imply that personalization should not be treated as a uniformly beneficial acceptance lever. In settings where objective criteria are salient and normatively legitimate, objective optimization may face lower initial resistance than preference-based personalization. This has direct implications for the design and communication of decision aids in finance and other high-stakes domains.

These implications extend beyond our experimental setting. Financial robo-advisors routinely face the choice between objective portfolio rules (e.g., mean-variance optimization) and preference-based personalization derived from risk profiling questionnaires (Capponi et al., 2022; Rühr, 2020). Our findings suggest that the very feature intended to increase subjective fit—tailoring the algorithm to the user’s profile—may inadvertently encroach on users’ sense of choice authority, deterring precisely those users who value exercising active control over their decisions. The same tension arises whenever algorithmic decision aids must be voluntarily adopted: in healthcare, treatment recommender systems can follow clinical-evidence guidelines or patient-preference protocols; in hiring, screening tools can apply standardized criteria or adaptively weight candidate profiles. In each case, the assumption that preference alignment promotes adoption deserves empirical scrutiny rather than being taken as a design axiom.

⁵Note that categorical non-delegation was not analyzed or reported in the earlier study, which is the reason for why we did not pre-register its analysis for the present study.

Future research should test mechanism-targeted interventions at first contact, including reframing of personalization logic, responsibility framing, and alternative explanation formats. It should also examine whether the non-delegator subgroup is stable across domains, stakes, and cultural contexts.

5.5 Contributions

This paper makes three contributions. First, it provides preregistered evidence that personalization can reduce, rather than increase, delegation at first contact. Second, it shows that resistance is not only transient: personalization is associated with substantially more categorical non-delegation. Third, by comparing two algorithmic decision rules within the same task, it clarifies that behavioral uptake depends on perceived rule characteristics, not only on whether advice is algorithmic or human. In doing so, the study also connects two literatures that are usually kept separate: the deterrent effect of preference-based personalization at adoption operates alongside a distinct, error-driven channel during use, whereby participants withdraw from delegation disproportionately after the algorithm produces a bad outcome, consistent with the literature on the lower forgiveness of algorithmic errors (Dietvorst et al., 2018; Renier et al., 2021).

5.6 Data and code availability

Data, analysis code, and study materials will be deposited in a public repository upon acceptance/publication of this manuscript.

5.7 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Opus 4.8 in order to assist in content organization as well as improving language and readability of both the manuscript and the analysis code (at the same time as allowing for an additional check on the statistical analyses). After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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